



The rock group Judas Priest was sued by the families of two teenage boys who committed suicide. The families claimed that subliminal messages on the rock group's album *Stained Class* prompted the boys to commit suicide. A witness for the families demonstrated that, if one song was filtered, slowed down, and played backward, people could hear the phrase "do it." The defense replied that the band had not intentionally inserted this backward message and that, even if they had, the message would not drive a person to suicide. The court found the band not guilty.

esteem" messages were labeled "memory tapes" and some tapes with "memory" messages were labeled "self-esteem tapes." After one month of listening, most who *thought* they were listening to self-esteem tapes said they had greatly improved their self-esteem; those who *thought* they were listening to memory tapes said their memory had greatly improved. What they were *actually* hearing made no difference. In other words, if people improved their memory—and some of them did improve, although not nearly as much as they thought they did—the improvement depended on their expectations, not on the tapes themselves (Greenwald, Spangenberg, Pratkanis, & Eskanazi, 1991).

What Subliminal Perception Can Do

We have just considered what subliminal messages *cannot* do. Now let's consider what they apparently *can* do:

If people are subliminally exposed to a simple picture and then asked to choose between that picture and another one (both now plainly visible), about 60–65% will choose the picture they had seen subliminally. Although this is not a very strong effect, it does last for a week or longer (Bornstein, 1989).

If people see a word subliminally, it influences their later perception of an easily visible stimulus (Dixon, 1981). For example, people watch a screen where the word PENCIL is flashed briefly in the midst of a cluttered background. Then they see a set of letters such as TERIW or WRITE and they are supposed to say, as quickly as possible, whether it is a word. People who have just seen the subliminal stimulus PENCIL respond a little quicker than usual that WRITE is a word.

A number of researchers have claimed that brief, subliminal exposure to an emotional message produces an emotional response. In one study, undergraduate students showed mild signs of nervousness or discomfort after viewing a very brief presentation of the message NO ONE LOVES ME but not after viewing the unemotional message NO ONE LIFTS IT (Masling, Bornstein, Poynton, Reid, & Katkin, 1991). This study was not set up to measure how long the effect might last. Other studies have reported fairly long-lasting emotional effects of subliminal stimuli (Hardaway, 1990), but such conclusions remain highly controversial.

The fact that subliminal perception can affect behavior at all is theoretically interesting. It shows that we are not consciously aware of all the information we process or all the events that influence us (Greenwald & Draine, 1997). However, the effects of subliminal perception are much smaller than some people hope and other people fear.

Perception and the Recognition of Patterns

Now let's consider how we perceive complex patterns. People are much better at recognizing complex patterns, especially faces, than at explaining how we do it. When people try to copy a photo of a face, most do a poor job, even though they could trace it reasonably accurately. The problem is not that they cannot control the pencil well enough; it is that they do not pay attention to all the details they need to copy the photo (Cohen & Bennett, 1997), even though they do register those details well enough to recognize the face.

When you attend your 25th high-school reunion, you will probably recognize many people, despite major changes in their appearance. Can you match the high-school photos in Figure 4.36 with the photos of the same people as they looked 25 years later? Probably not, but other people who had attended that high school succeeded with a respectable 49% accuracy (Bruck, Cavanaugh, & Ceci, 1991). They were able to focus on the invariant features that defined each person's face in spite of the changes that had occurred over the years.

But how do we recognize faces? To some extent we do so by unusual characteristics—a peculiar shape of nose, a characteristic mouth, or especially thick eyebrows. However, recognizing a whole face is apparently different in

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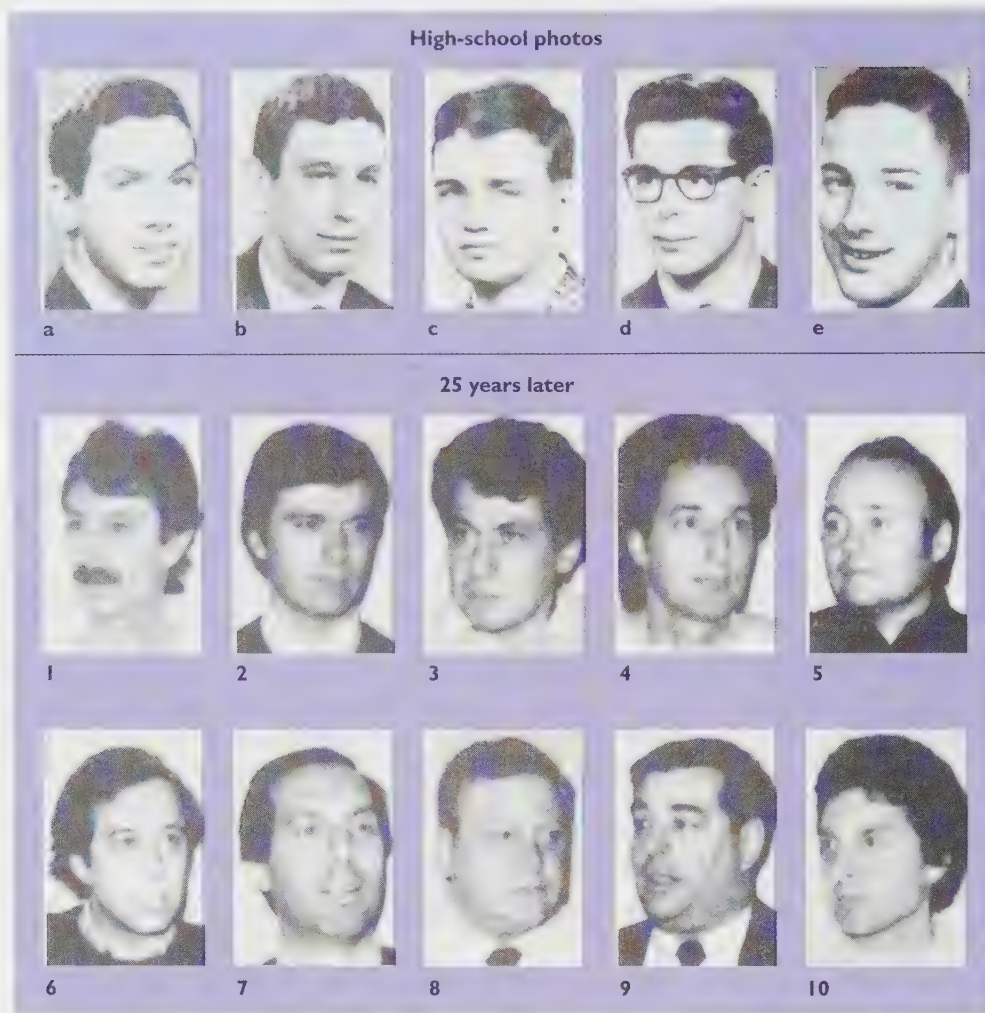


FIGURE 4.36 High-school photos and the same people 25 years later. Can you match the photos in the two sets? (Check answer C on page 152.) (From Bruck, Cavanaugh, & Ceci, 1991.)

some important ways from recognizing other kinds of objects, including pieces of a face. People with certain kinds of brain damage are unable to recognize faces, in spite of being able to read and recognize many kinds of objects (Farah, 1992). People with other kinds of brain damage can recognize faces normally but cannot recognize simple objects such as coffee cups or read even short words (Moscovitch, Winocur, & Behrmann, 1997). Particular parts of the cerebral cortex are more active when people look at faces than when they look at other complex patterns, such as flowers (McCarthy, Puce, Gore, & Allison, 1997), and many psychologists believe that the brain has a special “module” devoted specifically to face recognition.

Regardless of this module hypothesis, we do recognize faces as a whole, not just by specific features such as the eyes and nose. To illustrate, consider Figure 4.37. Presumably, you will identify the two faces immediately as those of U. S. President Bill Clinton and Vice President Al Gore. But now look more closely: The faces were digitally manipulated via computer so that they are exactly alike—about halfway between Clinton’s real face and Gore’s (Sinha & Poggio, 1996). You recognize the two faces partly by hair, head shape, relative height, and so forth. In short, visual recognition depends on both the isolated features and the overall context.



FIGURE 4.37 Can you easily recognize these faces as Bill Clinton and Al Gore? Note that the faces themselves have been altered to be identical. We recognize the individuals partly by facial features, but also partly by hair, head size, shape, and even the overall context of the picture.

The Feature-Detector Approach

Even explaining how we recognize a simple letter of the alphabet is difficult enough. According to one explanation, we begin recognition by breaking a complex stimulus into its component parts. For example, when we look at a letter of the alphabet, *specialized neurons* in the visual cortex, called **feature detectors**, *respond to the presence of certain simple features, such as lines and angles*. That is, one neuron in your visual cortex might become active only when you are looking at a horizontal line in a particular location. That feature detector would be detecting the feature “horizontal line.” Other neurons might detect horizontal lines in other locations, vertical lines, and so forth.

Feature detectors certainly cannot provide the whole explanation for how we perceive letters, much less faces. For example, we perceive the words in Figure 4.38a as CAT and HAT, even though the A in CAT is identical to the H in HAT, and therefore both of them stimulate the same feature detectors. Likewise, the character in the center of 4.38b can be read as either the letter B or the number 13. Feature detectors are essential in the early stages of visual



FIGURE 4.38 We perceive elements differently depending on their context. In (a), the A in CAT is the same as the H in HAT, but we perceive them differently. In (b), the central character can appear to be a B or the number 13, depending on whether we read horizontally or vertically. (Part b from Kim, 1989.)

perception, but the perception of a complex pattern requires more than just feature detectors.

Feature Detectors in the Human Visual System

We can easily imagine feature detectors in the human brain, and we can imagine all kinds of properties for them. But what evidence do we have for their existence? We have two kinds of evidence, one from laboratory animals and one from humans.

EXPERIMENT I

Hypothesis Neurons in the visual cortex of cats and monkeys will respond specifically when light strikes the retina in a particular pattern.

Method Two pioneers in the study of the visual cortex, David Hubel and Torsten Wiesel (1981 Nobel Prize winners in physiology and medicine), inserted thin electrodes into cells of the occipital cortex of cats and monkeys and then recorded the activity of those cells when various light patterns struck the animals' retinas. At first they used mere points of light; later they tried lines (Figure 4.39).

Results They found that each cell responds best in the presence of a particular stimulus (Hubel & Wiesel, 1968). Some cells become active only when a vertical bar of light strikes a given portion of the retina. Others become active only when a horizontal bar strikes the retina. In other words, such cells appear to act as feature detectors.

In later experiments, Hubel and Wiesel and other investigators found cells that respond to other kinds of features, such as movement in a particular direction.

Interpretation Hubel and Wiesel reported feature-detector neurons in both cats and monkeys. If the organization of

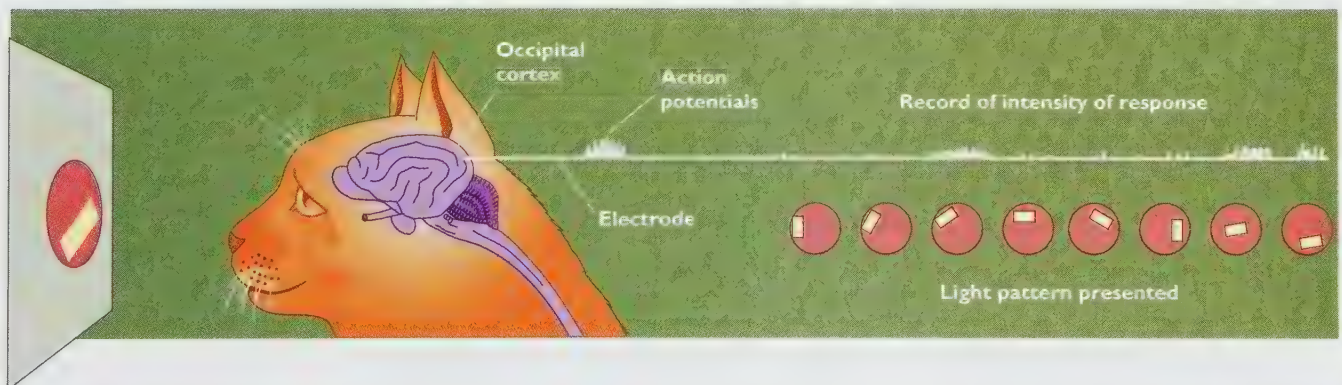


FIGURE 4.39 Hubel and Wiesel implanted electrodes to record the activity of neurons in the occipital cortex of a cat. Then they compared the responses evoked by various patterns of light and darkness on the retina. In most cases, a neuron responded vigorously when a portion of the retina saw a bar of light oriented at a particular angle. When the angle of the bar changed, that cell became silent but another cell responded.

the occipital cortex is similar in species as distantly related as cats and monkeys, it is likely (though not certain) to be similar in humans as well.

A second line of evidence is based on the following reasoning: If the human cortex does contain feature-detector cells, one type of cell should become fatigued after we stare for a time at the features that excite it. When we look away, we should see an aftereffect created by the inactivity of that type of cell. (Recall the negative afterimage in color vision, as shown by Figure 4.15.)

One example of this phenomenon is the **waterfall illusion**: *If you stare at a waterfall for a minute or more and then turn your eyes to some nearby cliffs, the cliffs will appear to flow upward.* By staring at the waterfall, you fatigue the neurons that respond to downward motion. When you look away, those neurons become inactive, but others that respond to upward motion continue their normal activity. Even though the motionless cliffs stimulate those neurons only weakly, the stimulation is enough to produce an illusion of upward motion.

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For another example, here is a demonstration that you can perform yourself.

EXPERIMENT 2

Hypothesis After you stare at one set of vertical lines, you will fatigue the feature detectors that respond to lines of a particular width. If you then look at lines slightly wider or narrower than the original ones, they will appear to be even wider or narrower than they really are.

Method Cover the right half of Figure 4.40 and stare at the little rectangle in the middle of the left half for at least 1 minute. (Staring for a longer time will increase the effect.) Do not stare at just one point; move your focus around within the rectangle. Then look at the square in the center of the right part of the figure, and compare the spacing between the lines of the top and bottom gratings (Blakemore & Sutton, 1969).

Results What did you perceive in the right half of the figure? People generally report that the top lines look narrower than they really are and that the bottom lines look wider.

Interpretation Staring at the left part of the figure fatigues one set of cells sensitive to wide lines in the top part of the figure and another set sensitive to narrow lines in the bottom part. Then, when you look at lines of medium width, the fatigued cells become inactive. Therefore, your perception is dominated by cells sensitive to narrower lines in the top part and to wider lines in the bottom part.

To summarize, we have two types of evidence for the existence of visual feature detectors: (1) The brains of other species contain cells with the properties of feature detectors, and (2) after staring at certain patterns, we see aftereffects that can be explained as fatigue of feature-detector cells in the brain.

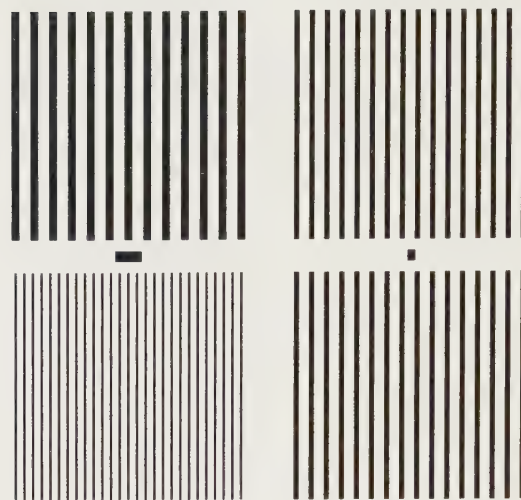
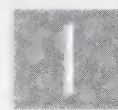
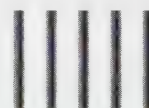


FIGURE 4.40 Use this display to fatigue your feature detectors and create an afterimage. Follow the directions in Experiment 2. (From Blakemore & Sutton, 1969.)

The research just described was only the start of an enormous amount of activity by laboratories throughout the world; later results have led to revised views of what the earlier results mean. For example, even though certain neurons respond better to a single vertical line



than to points or lines of other orientations, the vertical line may not be the best stimulus for exciting those neurons. Most respond even more strongly to a sine-wave grating of lines:



Thus the feature that such cells detect is probably more complex than just a line. Furthermore, because each cell responds to stimuli as different as a line and a group of lines, obviously no one cell provides an unambiguous message about what someone is seeing at any moment. The perception emerges from the pattern of activity of a large population of neurons (Hughes, Nozawa, & Kitterle, 1996).

One important point about scientific advances: A single line of evidence—even excellent, Nobel Prize-winning evidence—seldom provides the final answer to any question. We should always look for multiple ways to test a hypothesis; even if several kinds of evidence support a conclusion, a great many unanswered questions can still remain.



Do Feature Detectors Explain Perception?

The neurons I have been describing are active during the early stages of visual processing. Do we simply add up the responses of a great many feature detectors, so that the sum of enough feature detectors constitutes your perception of, say, your psychology professor's face? No, brain researchers doubt that the responses of simple horizontal- or vertical-line feature detectors equal our conscious visual experience (He, Cavanagh, & Intiligator, 1996). Rather, they are just a preliminary step in a complex set of visual processes occupying most of the brain.

To illustrate why simple feature detectors alone cannot explain perception, consider Figure 4.41 (based on Kanizsa, 1979). Parts a and b are composed of small geometric forms. Although we might guess that part a is made up of segments of a three-dimensional cube, we do not *see* the cube. Part b hardly even suggests a cube. In parts c and d, the added lines provide a context that enables us to see the cube. In part e, the deletion of short lines from a enables us to *see* imaginary lines that provide the same context. In c, d, and e, we have perceptually organized a meaningful pattern that goes well beyond the sum of the individual lines, taken one at a time.

Similarly, in Figure 4.42a we see a series of meaningless patches. In Figure 4.42b, the addition of some black glop immediately enables us to perceive these same patches as the word *psychology* (Bregman, 1981). We can perceive the letters in part b only by imposing an active interpretation on the pattern.

The Gestalt Psychology Approach

Figure 4.43, which we see as the overall shape of an airplane, is a photo of several hundred people. Out of context, one person is no more a piece of an airplane than a piece of anything else; the plane is the overall pattern, not the sum of the parts. Recall also Figure 4.32 from earlier in this chapter: The photograph is composed entirely of dots, but we perceive a pattern of objects, not just a collection of dots.

Such observations derive from **Gestalt psychology**, a field that focuses on our ability to perceive overall patterns.

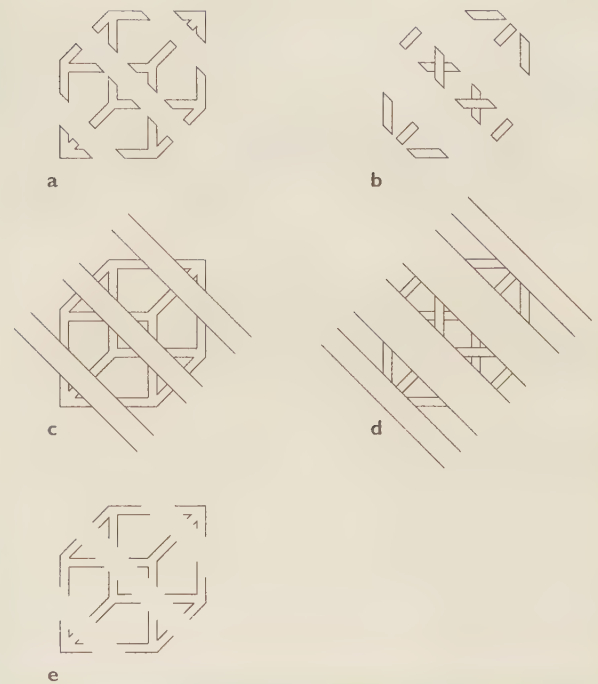


FIGURE 4.41 This picture is a puzzle until a context is introduced. Then a cube “emerges” from meaningless lines. (From Kanizsa, 1979.)

Gestalt (geh-SHTALT) is a German word; there is no exact English equivalent, although *configuration* and *pattern* come close. The founders of Gestalt psychology rejected the idea that a perception can be broken down into its component parts. If a melody is broken up into individual notes, the melody is lost. Their slogan was, “The whole is different from the sum of its parts.”

According to Gestalt psychologists, visual perception is an active creation, not just the adding up of lines, dots, or other pieces. We considered examples of this principle in Figures 4.40 and 4.41. Here are some further examples:

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Figure 4.44 shows a photo and a drawing of two animals. As you look at these pictures, you may see the animals almost at once, or you may see

FIGURE 4.42 Someone familiar with only the Hebrew or Arabic alphabet would not see meaningful patterns emerge in part (b). (After Bregman, 1981.)

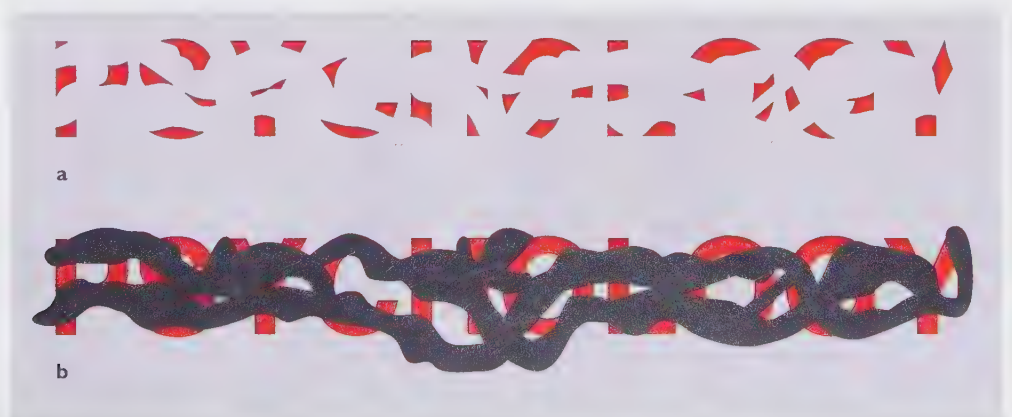




FIGURE 4.43 According to Gestalt psychology, the whole is different from the sum of its parts. Here, we perceive an assembly of several hundred people as an airplane.

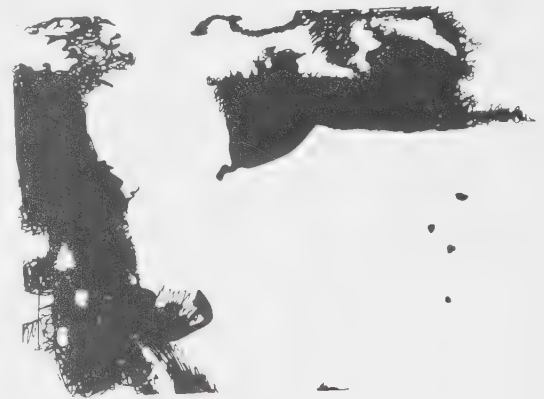
meaningless black and white patches from which you suddenly organize the perception of animals. (If you give up, check answer D, page 152.) To perceive the animals, you must separate **figure and ground**—that is, you must distinguish the *object from the background*. Ordinarily, you make that distinction almost instantly; only in special cases like this one do you become aware of the process.

Figure 4.45 contains five **reversible figures**, *stimuli that can be perceived in more than one way*. In effect, we test hypotheses: “Is this the front of the object or is that the front? Is the object facing left or right? Is this section the foreground or the background?” In Figure 4.45, part a is called the *Necker cube*, after the psychologist who first called attention to it. Which is the front face of the cube? If you look long enough, you will see it two ways. You can see part b either as a vase or as two profiles. In part c, with a little imagination, you might see a woman’s face or a man blowing a horn. (If you need help, check answer E on page 152.) Part d (from Boring, 1930) shows both an old woman and a young woman. Almost everyone sees one or the other immediately, but many people lock into one perception so tightly that they cannot see the other one. Part e was drawn by an 8-year-old girl who intended it as the picture of a face. Can you find another possibility? (If you have trouble with parts d or e, check answers F and G, page 152.) Overall, the point of the reversible figures is that we perceive by imposing order on an array, not just by adding up lines and points.

The Gestalt psychologists described some principles of how we organize perceptions into meaningful wholes, as illustrated in Figure 4.46. **Proximity** is the *tendency to perceive objects that are close together as belonging to a group*. The objects in part a form two groups because of

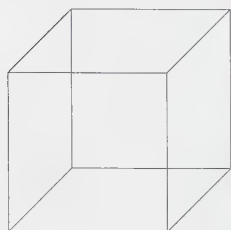


a



b

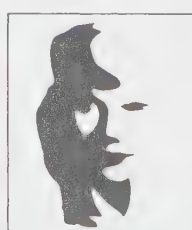
FIGURE 4.44 Do you see an animal in each picture? If not, check answer D, page 152. (Part b from Dallenbach, 1951.)



a



b



c



d



e

FIGURE 4.45 Reversible figures: (a) The Necker cube. Which is the front face? (b) Faces or a vase. (c) Sax player and woman’s face (“Sara Nader”). (d) An old woman and a young woman. (e) A face or what? (Part d from Boring, 1930.)

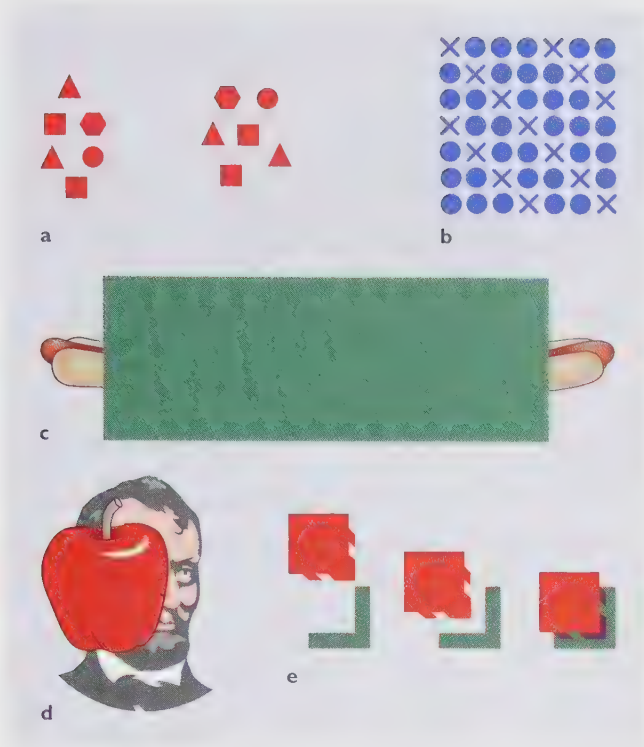
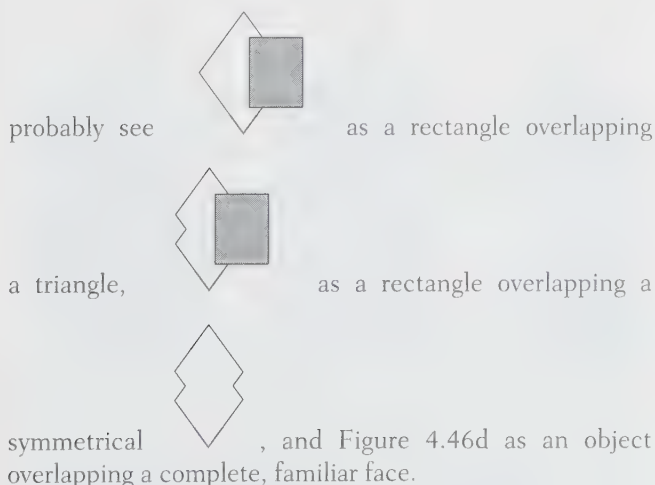


FIGURE 4.46 Gestalt principles of (a) proximity, (b) similarity, (c) continuation, (d) closure, and (e) good figure.

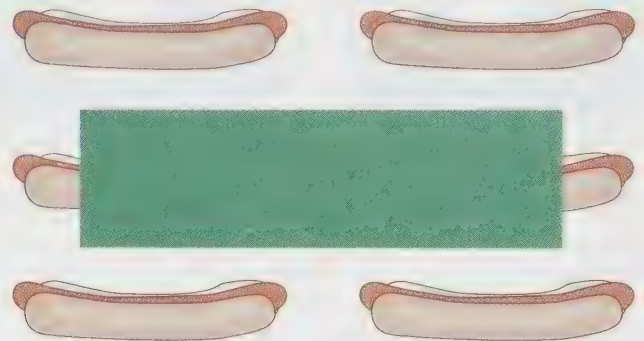
their proximity. The *tendency to perceive objects that resemble each other as forming a group* is called **similarity**. The objects in b group into Xs and Os because of similarity. When lines are interrupted, as in c, we may perceive **continuation**, a *filling in of the gaps*. You probably perceive this illustration as a rectangle covering the center of one very elongated hot dog.

When a familiar figure is interrupted, we perceive a **closure** of the figure—that is, *we imagine the rest of the figure*. The figure we imagine completes what we already see in a way that is simple, symmetrical, or consistent with our past experience (Shimaya, 1997). For example, you



probably see  as a rectangle overlapping a triangle,  as a rectangle overlapping a symmetrical , and Figure 4.46d as an object overlapping a complete, familiar face.

takes into account more than just a continuation of the lines. For example, in Figure 4.46c, you fill in the gaps to perceive one long hot dog. With some additional context, you would probably perceive the same pattern as two shorter hot dogs:



Finally, we tend to perceive a **good figure**—a *simple, symmetrical figure*. As a rule, simple and symmetrical figures are more likely to occur in nature than irregular figures.

If you see , it is more likely to be composed of one ellipse overlapping another than to be

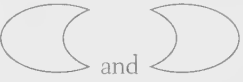
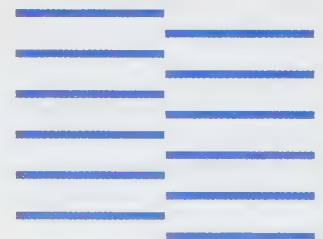
composed of  meeting at exactly the right point. In Figure 4.46e, even after we see that the right-hand drawing is a green backward L overlapping part of an irregular object, we continue to perceive it as a red square overlapping a green square. In Figure 4.47, we perceive two striped surfaces, either adjoining side by side or one overlapping the other. That perception is so convincing that you may have to look carefully to persuade yourself that there is no vertical line down the center of the drawing.

Figure 4.48 demonstrates that the principle of good figure does not always lead us to perceive the most likely stimulus (Grossberg, 1997). Most people see part a as a large white cross overlapping a thinner white square, although it is also possible (and equally logical) to see a thin square overlapping a large cross. In parts b and c, we see the thin square clearly superimposed on the left and right arms of the cross; nevertheless, at the top and bottom we tend to see the large cross overlapping the thin square. That is, most people see the square as looping over and

FIGURE 4.47 Two sets of stripes generate an illusion of a vertical line down the center, separating one set of stripes from the other.



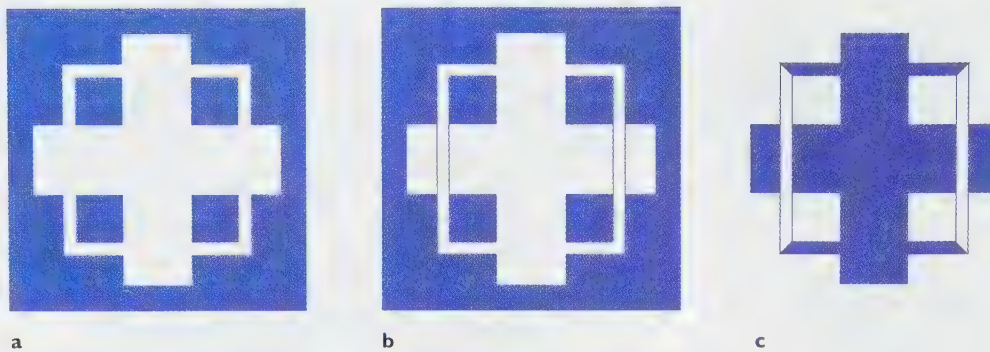


FIGURE 4.48 Do you see a thick cross on top of a thin square, or a square on top of a cross? In part a, both are equally possible, but most people perceive the cross as being on top. In parts b and c, we are shown that the square is on top of the left and right arms of the cross, but most people still see the top and bottom arms of the cross as lying above the square. (Modified from Petter, 1956; and Kanizsa, 1985.)

under the cross, instead of the more likely event that it lies above the cross in all locations.

Perceiving one pattern as lying “on top of” another can have important consequences. Consider Figure 4.49. The dark bars on the right appear to lie on top of a pink rectangle, and therefore the pink rectangle is on a white surface. The contrast between the pink and the white makes the pink appear darker. Meanwhile, the pink on the left appears to lie above the dark bars, making it appear very light by contrast. Overall, the pink bars on the right look much darker than those on the left, even though they are actually the same shade of pink.

CONCEPT CHECK

- 15.** Which of the Gestalt principles were operating in your perception of Figures 4.41 and 4.42? (Check your answers on page 152.)

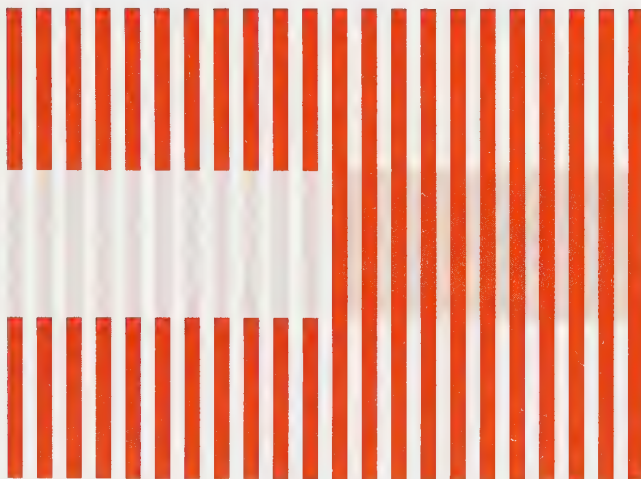


FIGURE 4.49 Because pink bars appear to lie above the dark bars on the left and below them on the right, we see a contrast between pink and dark red on the left, pink and white on the right. Therefore, we see the pink bars on the right as darker, even though they are actually the same shade as the others.

Gestalt Principles in Hearing

The perceptual organization principles of Gestalt psychology apply to hearing as well as to vision. There are reversible figures in sound, just as there are in vision. For instance, you can hear the sound of a clock as “tick, tock, tick, tock” or as “tock, tick, tock, tick.” You can hear your windshield wipers going “dunga, dunga” or “gadung, gadung.”

As with visual reversible figures, people occasionally get so locked into one interpretation of something they hear that they have trouble hearing it any other way. For example,

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read this sentence to a friend: “The matadors fish on Friday.” Pause long enough to make sure your friend has understood the sentence. Then say: “The cat on the mat adores fish on Friday.” If you read the second sentence normally, without pausing between mat and adores, your friend is likely to be puzzled: “Huh? The cat on the matadors . . .?” Had you not just read the first sentence, your friend would have easily understood the second sentence.

Feature Detectors and Gestalt Psychology

The Gestalt approach to perception does not conflict with the feature-detector approach as much as it might seem. The feature-detector approach describes the first stages of perception—how the brain takes individual points of light and connects them into lines and then into more complex features. According to the feature-detector approach, the brain says, “I see these points here, here, and here, so there must be a line. I see a line here and another line connecting with it here, so there must be a letter L.”

The Gestalt approach describes how we combine visual input with our knowledge and expectations. According to the Gestalt interpretation, the brain says, “I see what looks like a circle, so the missing piece must be part of a circle too.”

Which view is correct? Both, of course. Our perception must assemble the individual points of light or bits of

sound, but once it forms a tentative interpretation of the pattern, it uses that interpretation to organize or reorganize the information.

Perception of Movement and Depth

As an automobile drives away from us, its image on the retina grows smaller, yet we perceive it as moving, not as shrinking. That perception illustrates **visual constancy**—our *tendency to perceive objects as keeping their shape, size, and color, even though what actually strikes our retina may be changing from time to time*. Figure 4.50 shows examples of two visual constancies: shape constancy and size constancy. Constancies depend on our familiarity with objects and on our ability to estimate distances and angles of view. For example, we know that a door is still rectangular even when we view it from an odd angle. But to recognize that an object keeps its shape and size, we have to perceive movement or changes in distance or angle. How do we do so?

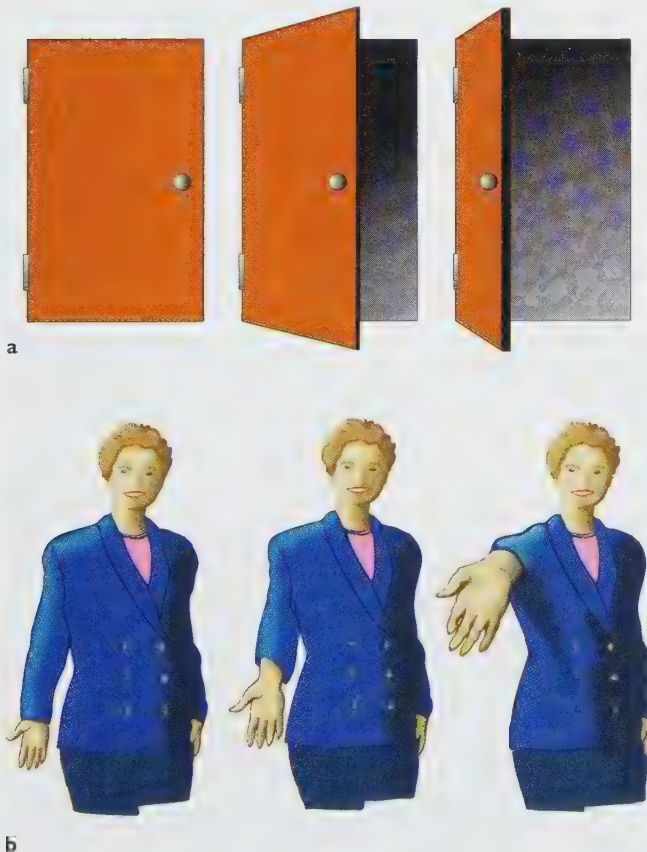


FIGURE 4.50 (a) Shape constancy: We perceive all three doors as rectangles. (b) Size constancy: We perceive all three hands as equal in size.

Perception of Movement

It is common sense to assume that anyone who can see a rabbit should be able to see its size, shape, and color, and its direction and speed of movement. In this case, our common sense is wrong. You already know that some people are color-blind and therefore cannot see the rabbit's color. Also, as mentioned briefly in Chapter 3, some people are motion-blind and therefore their ability to see the rabbit's movement is impaired. Motionblindness results from damage to a small area in the temporal lobe of the cortex (Zihl, von Cramon, & Mai, 1983). This rare condition illustrates a major point: The visual system of the brain has separate pathways to analyze different aspects of what we see. One pathway analyzes shape, another analyzes color, and yet another analyzes movement (Zeki, 1993).

The detection of motion in the visual world raises some interesting issues, including how we distinguish between our own movement and the movement of objects. Try this simple demonstration: Hold an object in front of your eyes and then move it to the right. Now hold the object in front of your eyes and move your eyes to the left. The image of the object moves across your retina in the same way, regardless of whether you move the object or move your eyes. Yet, you perceive the object as moving in one case but not in the other. Why?

The object does not appear to move when you move your eyes for two reasons. One reason is that the vestibular system constantly keeps the visual areas of the brain informed of movements of your head. When your brain knows that your eyes have moved to the left, it interprets a change in what you see as being a result of that movement. One man with a rare kind of brain damage was unable to connect his eye movements with his perceptions. Whenever he moved his eyes, he perceived the world as moving. Whenever he watched moving objects or watched anything while he himself moved, he became dizzy and nauseated (Haarmeier, Thier, Repnow, & Petersen, 1997).

The second reason that the object does not appear to move is that we perceive motion when an object moves *relative to the background* (Gibson, 1968). For example, when you walk forward, stationary objects in your environment move across your retina. If something fails to move across your retina, you perceive it as moving in the same direction as you are.

What do we perceive when an object is stationary and the background is moving? That seldom happens, but when it does, we may *incorrectly perceive the object as moving against a stationary background*, a phenomenon called **induced movement**. For example, when you watch clouds moving slowly across the moon from left to right, you generally perceive the clouds as a stationary background and the moon as an object moving from right to left. Induced movement is a form of *apparent movement*, as opposed to *real movement*.

I have already mentioned the waterfall illusion (page 138) as an example of apparent movement. Another example is **stroboscopic movement**, an *illusion of movement created by a rapid succession of stationary images*. When a scene is flashed on a screen and is followed a split second later by a second scene slightly different from the first, you perceive the objects as having moved smoothly from their location in the first scene to their location in the second scene (Figure 4.51). Motion pictures are actually a series of still photos flashed on the screen.

We also experience an *illusion of movement created when two or more stationary lights separated by a short distance blink on and off at regular intervals*. Your brain creates the sense of motion in what is called the **phi effect**. You may have noticed signs in front of restaurants or motels that make use of this effect. As the lights blink on and off, the arrow seems to be moving and inviting you to come inside.

Our ability to detect visual movement played an interesting role in the history of astronomy. In 1930, Clyde Tombaugh was searching the skies for a possible undiscovered planet beyond Neptune. He photographed each region of the sky twice, several days apart. A planet, unlike a star, would move from one photo to the next. However, how would he find one tiny dot that moved, among all the countless unmoving dots in the sky? He put each pair of photos on a machine that would flip back and forth between one photo and the other. When he came to the correct pair of photos, the machine flipped back and forth between them, and he immediately noticed the one moving dot (Tombaugh, 1980). We now know that little dot as the planet Pluto (Figure 4.52).

Depth Perception

Although we live in a world of three dimensions, our retinas are in effect two-dimensional surfaces. **Depth perception**, our *perception of distance*, enables us to experience the world in three dimensions. This perception depends on several factors.

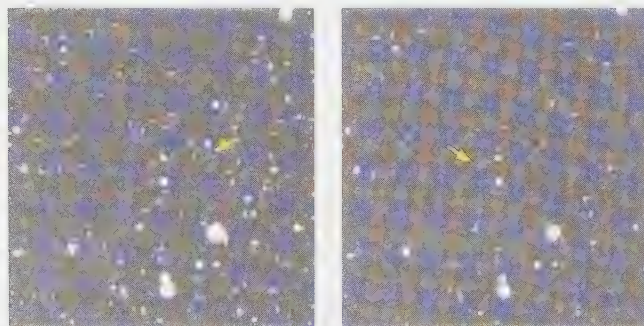


FIGURE 4.52 Clyde Tombaugh used movement perception to discover the planet Pluto. He photographed each area of the sky twice, several days apart. Then he used a machine to flip back and forth between the two photos of each pair. When he came to one part of the sky, he immediately noticed one dot that moved between the two photos. That dot was the planet Pluto.



One factor is **retinal disparity**—*the difference in the apparent position of an object as seen by the left and right retinas*. Try this: Hold one finger at arm's length. Focus on it with one eye and then with the other. Note that the apparent position of your finger shifts with respect to the background. Now hold your finger closer to your face and repeat the experiment. Notice that the apparent position of your finger shifts even more. The discrepancy between the slightly different views the two eyes see becomes greater as the object comes closer. We use the amount of discrepancy to gauge distance.

A second cue for depth perception is the **convergence** of our eyes—that is, the *degree to which they turn in to focus on a close object* (Figure 4.53). When you focus on a distant object, your eyes are looking in almost parallel directions. When you focus on something close, your eyes turn in; you can sense the tension of your eye muscles. The more the muscles pull, the closer the object must be.

Retinal disparity and convergence are called **binocular cues**, because they *depend on the action of both eyes*. **Monocular cues** enable a person to *judge depth and distance effectively with just one eye*, or when both eyes see

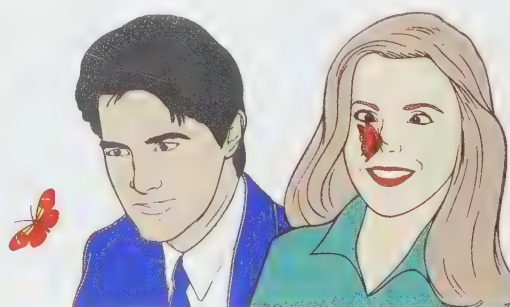


FIGURE 4.53 Convergence of the eyes as a cue to distance. The more this viewer must converge her eyes toward each other in order to focus on an object, the closer the object must be.



FIGURE 4.51 When you watch a movie, you are unaware of the thousands of still photographs flickering by at a rate of 86,400 per hour. The sequence of photographs printed here conveys a sense of motion in another way.

the same image, as they do when you look at a picture. Several monocular cues help us judge the approximate distance of objects, such as those in Figure 4.54.

Object size: Other things being equal, an object close to us produces a larger image than does one that is farther away. This cue is useful only if we already know the approximate actual size of the objects. For example, the roller skater in the photo produces a larger image than does the parked van, which we know is actually larger. So we see the skater as closer. The rocks in the sea, however, are not of equal size in reality, so the relative sizes of their images do not tell us anything about their distance.

Linear perspective: As parallel lines stretch out toward the horizon, they come closer and closer together. Examine the road in Figure 4.54. At the bottom of the photo (close to the viewer), the edges of the road are far apart; at greater distances,

the edges of the road come closer together. The closer the lines come, the more distant we perceive them to be.

Detail: We see nearby objects, such as the roller skater, in much detail. More distant objects are increasingly hazy and less detailed.

Interposition: A nearby object interrupts our view of a more distant object. Interposition is our surest way of seeing which rocks are closer than others.

Texture gradient: Notice the posts on the safety rail on the right side of the road. At greater distances, the posts come closer and closer together. The “packed together” appearance of objects gives us another cue to their approximate distance.

Shadows: Shadows are hardly prominent in Figure 4.54, but when present they can provide another cue to help us interpret the sizes and positions of objects in a picture.



FIGURE 4.54 We can judge depth and distance in a photograph, using certain useful cues, even if we use only one eye. (1) Closer objects occupy more space on the retina (or in the photograph) than do distant objects of the same type. (2) Nearer objects show more detail. (3) Closer objects overlap certain distant objects. (4) Objects in the foreground look sharper than objects do on the horizon. These are known as monocular cues.

Accommodation: The lens of the eye *accommodates*—that is, it changes shape—to focus on nearby objects, and your brain detects that change and thereby infers the distance to an object. Accommodation could help tell you how far away the photograph itself is, although it provides no information about the relative distances of objects in the photograph.

The ability to use these monocular cues to interpret an illustration depends partly on our experience with photographs and drawings. For example, in the two drawings of Figure 4.55, does it appear to you that the hunter is aiming his spear at the deer? When these drawings were shown to people of certain African cultures that create much sculpture but little drawing, many people said the hunter was aiming at a baby elephant (Hudson, 1960). This comparison illustrates that people have to learn how to use monocular cues to judge depth in drawings.

Another monocular cue helps us to perceive depth when we are looking at a three-dimensional scene, though it does not help us when we are looking at a photograph. When we are moving—riding along in a car, for example—close objects seem to pass by swiftly, although distant objects seem to pass by very slowly. *The faster an object passes by, the closer it must be.* That principle is **motion parallax**.

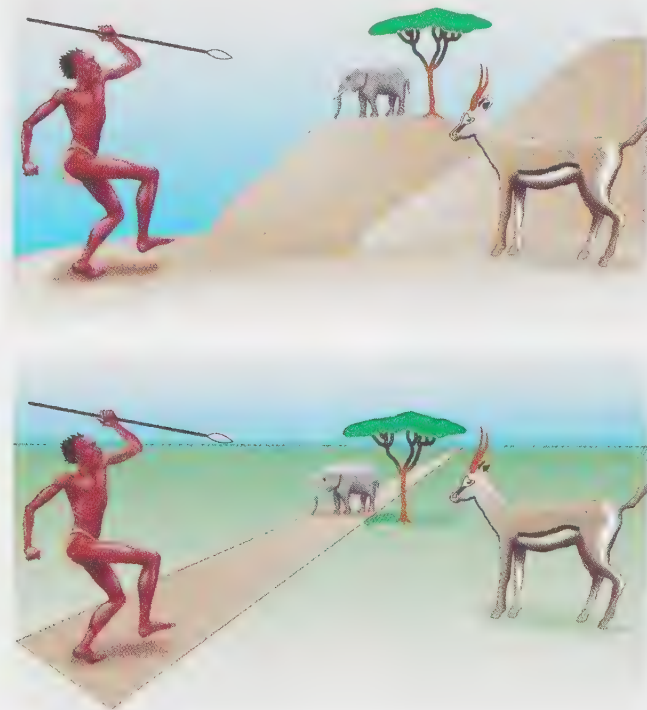


FIGURE 4.55 Which animal is the hunter attacking? Most readers of this text, using monocular cues to distance, will reply that the hunter is attacking the deer. However, many African subjects thought he was attacking “a baby elephant.” Evidently, the tendency to use monocular cues to distance depends on experience with photographs and drawings; the African subjects were from cultures with little such experience. (From Hudson, 1960.)



If you were a passenger on this train, the ground beside the tracks would appear to pass by more quickly than the more distant elements in the landscape. In this photo's version of motion parallax, the ground is blurred and more distant objects are crisp.

CONCEPT CHECKS

16. Which monocular cues to depth are available in Figure 4.55?
17. With three-dimensional photography, cameras take two views of the same scene from different locations through lenses with different color filters or with different polarized-light filters. The two views are then superimposed. The viewer looks at the composite view through special glasses so that one eye sees the view taken with one camera and the other eye sees the view taken with the other camera. Which depth cue is at work here? (Check your answers on page 152.)

Optical Illusions

Many people claim to have seen ghosts, flying saucers, the Loch Ness monster, Bigfoot, Santa's elves, or people floating in the air. Maybe they are lying, maybe they did see something extraordinary, or maybe they saw something ordinary but misinterpreted it. An **optical illusion** is a mis-

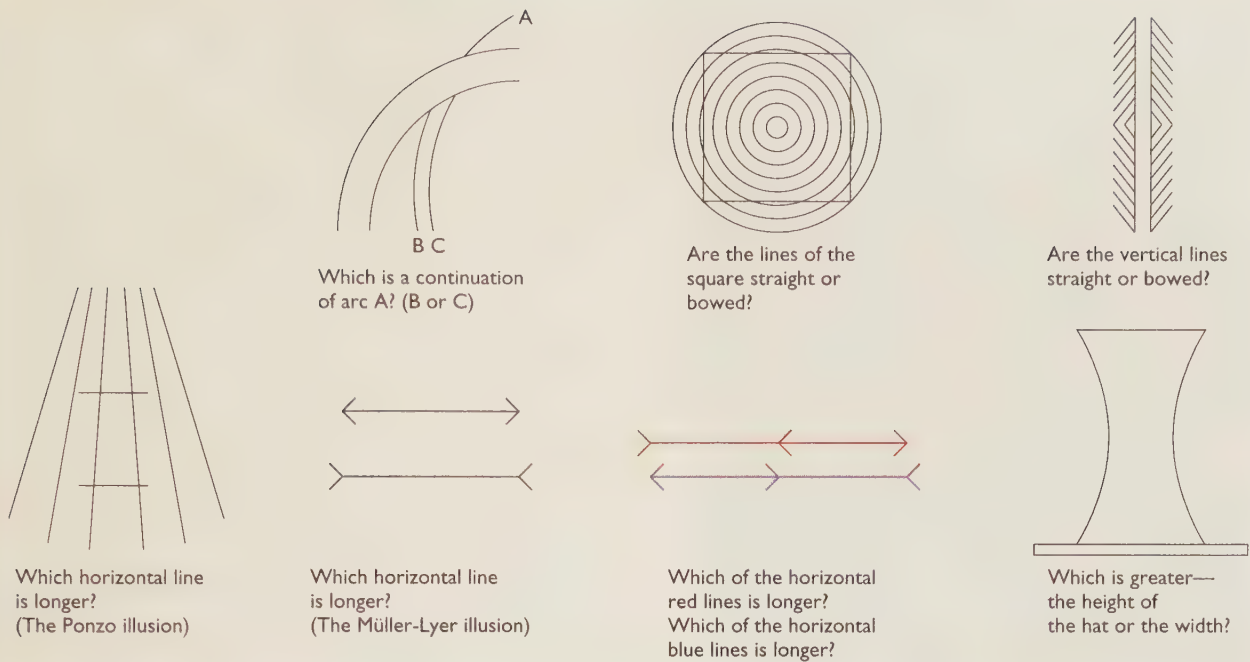


FIGURE 4.56 Many paintings rely on an optical illusion, but we are more aware of the illusion in geometric figures. (Check your answers with a ruler and a compass.)

interpretation of a visual stimulus. Figure 4.56 shows a few examples. Psychologists would like to develop a single explanation for all optical illusions. (Remember the principle of parsimony from Chapter 2.) Although psychologists have not fully succeeded, today we can explain a fair number of optical illusions based on the relationship between size perception and depth perception.

The Relationship Between Depth Perception and Size Perception

If you can estimate the size of an object, you can deduce its distance. If you can estimate its distance, you can deduce its size. Figure 4.57 shows that a given image on the retina may represent either a small, close object or a large, distant object. Watch what happens when you take a single image and change its apparent distance: Stare at Figure 4.15 again to form a negative af-

terimage. First examine the afterimage while you are looking at the wall across the room. Your afterimage will look fairly large. Then look at the afterimage against the palm of your hand. Suddenly, the image becomes very small. Move your hand backward and forward; you can make the apparent size grow and then shrink.

In the real world, we seldom have trouble estimating the size and distance of objects. When you walk along the street, for instance, you never wonder whether the people you see are giants one kilometer away or miniature people a few centimeters away. However, when you have fewer cues about the size or distance of an object, you can become confused (Figure 4.58). I once saw an airplane overhead and was unsure whether it was a small, remote-controlled toy airplane or a distant, full-size airplane. Airplanes come in many sizes, and the sky has few cues to distance.

A similar issue arises in reported sightings of UFOs. When people see an unfamiliar object in the sky, they can

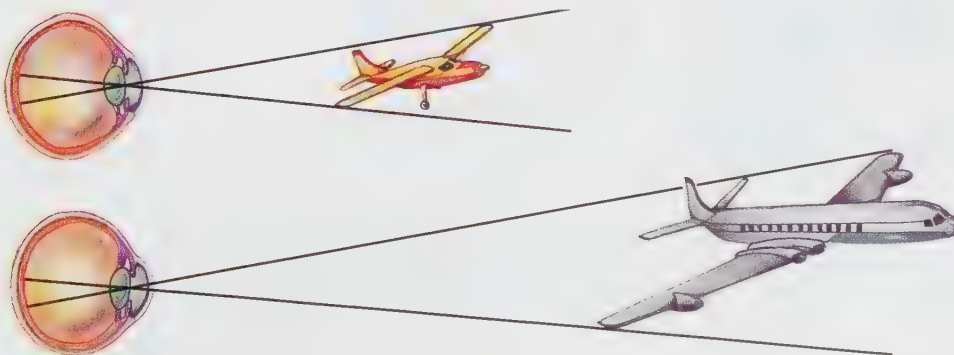


FIGURE 4.57 The trade-off between size and distance: A given image on the retina can indicate either a small, close object or a large, distant object.



FIGURE 4.58 Because fish come in many sizes, we can estimate the size of a fish only if we know how far away it is or if we can compare its size to other nearby objects. See what happens when you cover the man and then cover the hand.

easily misjudge its distance. If they overestimate its distance, they also will overestimate its size and speed.

What does all this have to do with optical illusions? Whenever we misjudge distance, we are likely to misjudge size as well. For example, Figure 4.59a shows people in the

Ames room (named for its designer, Adelbert Ames). The room is designed to look like a normal rectangular room, though its true dimensions are as shown in Figure 4.59b. The right corner is much closer than the left corner. The two young women are actually the same height. If we eliminated all the background cues, then we would correctly perceive the women as being the same size but at different distances. However, the apparently rectangular room provides such powerful (though misleading) cues to distance that the women appear to differ greatly in height.

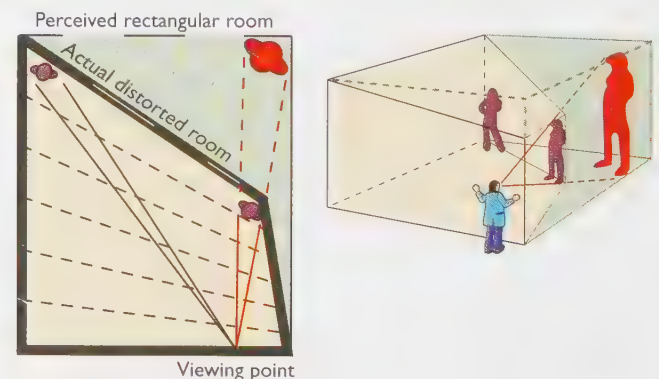
Even a two-dimensional drawing on a flat surface can offer cues that lead to erroneous depth perception. People who have had much experience with photos and drawings tend to interpret two-dimensional drawings as if they were three-dimensional. Figure 4.60 shows a bewildering two-prong/three-prong device and a round staircase that seems to run uphill all the way clockwise or downhill all the way counterclockwise. Both drawings puzzle us because we try to interpret them as three-dimensional objects.

In Figure 4.61a, we interpret the railroad track as heading into the distance. Similarly, because the background cues in part b suggest that the upper line is farther away than the lower line, we perceive the upper line as being larger. The same is true of the right-hand cylinder in part c. Recall from Figure 4.57 that when two objects produce the same-size image on the retina, we perceive the more distant one as being larger. In short, by perceiving two-dimensional representations as if they were three-dimensional, we misjudge distance and consequently misjudge size. When we are somehow misled by the cues that ordinarily ensure constancy in size and shape, we end up experiencing an optical illusion (Day, 1972).

We can experience an *auditory illusion* by a similar principle: If we misestimate the distance to a sound source, we will misestimate the intensity of the sound. In one study, experimenters misled students about the distance of a sound



a



b

FIGURE 4.59 The Ames room is a study in deceptive perception, designed to be viewed through a peephole with one eye. (a) Both of these people are actually the same height. We are so accustomed to rooms with right angles that we can't imagine how this apparently ordinary room creates this optical illusion. (b) This diagram shows the positions of the people in the Ames room and demonstrates how the illusion of distance is created. [Part (b) from Wilson et al., 1964.]

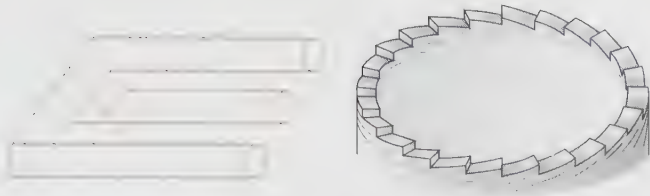


FIGURE 4.60 Expectations shape perceptions. These two-dimensional drawings puzzle us because we try to interpret them as three-dimensional objects.

by using the **visual capture effect**, the *tendency to hear a sound as coming from a visually prominent source*. (You experience this effect when you “hear” a voice coming from a ventriloquist’s dummy or from a movie or television screen.) The experimenters had an unchanging sound source that the students never saw, plus a silent “dummy loudspeaker” that moved. The students always thought they heard the sound coming from the dummy loudspeaker, regardless of where the experimenters placed it. When the loudspeaker was far away, most students said the sound was louder than when the speaker was close (Mershon, Desaulniers, Kiefer, Amerson, & Mills, 1981). Remember, the actual sound was the same in all cases. When people thought they heard such a sound from a greater distance, they interpreted it as being more intense.

Cross-Cultural Differences in Seeing Optical Illusions

For most of us, the Müller-Lyer illusion is one of the most convincing. (See Figure 4.62.) According to one attempt at explaining this illusion, the lines with outward-facing arrowheads (on the left in Figure 4.62) appear larger because they resemble the edges of the back of a building; the lines with inward-facing arrowheads resemble the front of a building. (See Figure 4.63.) If we perceive one line (a) as being closer than the other line (b), we are likely to interpret the closer one as shorter.

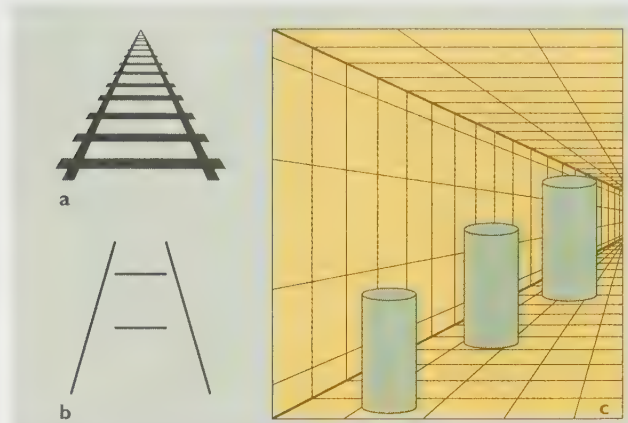
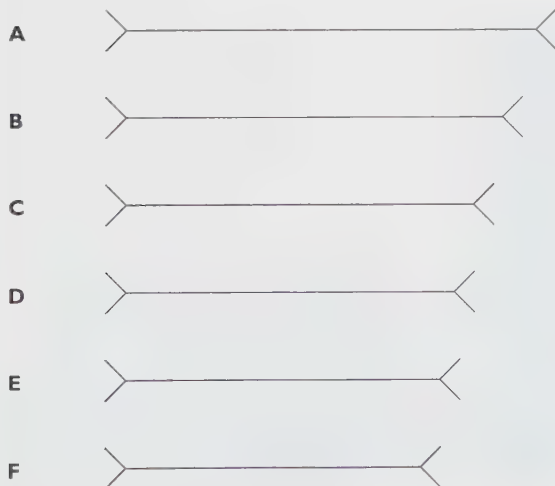


FIGURE 4.61 Some optical illusions depend on misjudgment of distances. In part (b), the top line looks longer because the perspective, resembling the railroad tracks in part (a), suggests a difference in distance. In part (c), the jar on the right seems larger because the context makes it appear farther away.

For several reasons, this explanation is not altogether convincing, but it did lead to the interesting prediction that the Müller-Lyer illusion might be stronger in cultures that have experience with rooms and buildings and weaker in cultures that build circular huts or that live in rain forests or other rural environments. The illusion might also depend on people’s experience with drawings of objects.

To test this hypothesis, researchers showed subjects stimuli similar to those in Figure 4.62 and determined which of the horizontal lines with outward arrows appeared the same length as the line with inward arrows. They found that the illusion was stronger for people who lived in cities and weaker for African bushmen (Segall, Campbell, & Herskovits, 1966) and Zambian farm-dwellers (Stewart, 1973). Even the bushmen and farm-dwellers did see the illusion, however; the difference was merely how large an illusion they saw. Furthermore, children generally experienced a larger illusion than adults did, indicating that the ability to see the illusion does not require extensive visual experience.

FIGURE 4.62 The Müller-Lyer illusion: Ignoring (or trying to ignore) the arrowheads, which of the horizontal lines on the left is the same length as the horizontal line at the right? (Check answer H on page 152.) For most people in the U.S., Canada, and Europe, this is a strong and convincing illusion. The illusion is present but apparently weaker for many people from less technological societies.

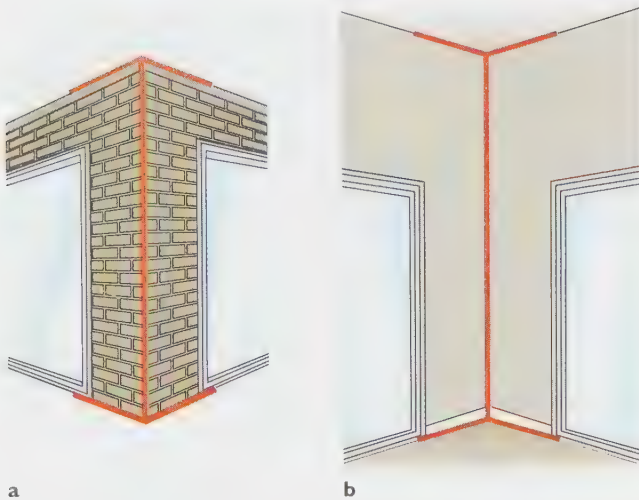


FIGURE 4.63 According to one interpretation of the Müller-Lyer illusion, inward-facing arrowheads make a line appear shorter because they make the line resemble the front of a building (as in a), but outward-facing arrowheads make the line resemble the back of a building (as in b). If we interpret the line with inward-facing arrowheads as closer, we will also interpret the line as shorter.

Clearly the results indicate a cultural influence, but they do not necessarily indicate a cultural difference in depth perception. All the testing was done with drawings on paper, and the differences among cultures probably reflect differences in experience with interpreting two-dimensional drawings (Montello, 1995).

The Moon Illusion

To most people, the *moon close to the horizon appears about 30% larger than it appears when it is higher in the sky*. This **moon illusion** is so convincing that some people have tried to explain it by referring to the bending of light rays by the atmosphere or another physical phenomenon. The explanation, however, must depend on the observer, not the light rays. If you actually measure the

moon image with navigational or photographic equipment, you will find that it is the same size at the horizon as it is higher in the sky. For example, Figure 4.64 shows the moon at two positions in the sky; you can measure the two images to demonstrate that they are really the same size. (The atmosphere's bending of light rays makes the moon look orange near the horizon, but it does not increase the size of the image.) However, photographs cannot capture the full strength of the moon illusion as we see it in real life. In Figure 4.64 (or any similar pair of photos), the moon looks almost the same at each position; in the actual night sky, though, the moon looks enormous at the horizon.

One possible explanation is that the vast terrain between the viewer and the horizon provides a basis for size comparison. When you see the moon at the horizon, you can compare it to the other objects you see at the horizon, all of which look tiny. By contrast, the moon looks large. When you see the moon high in the sky, however, it is surrounded only by the vast, featureless sky, so in contrast the moon appears relatively small (Baird, 1982; Restle, 1970).

A second possible explanation is that the terrain between the viewer and the horizon gives an impression of great distance. When the moon is high in the sky, we have no basis to judge distance, and perhaps we thus unconsciously see the overhead moon as closer than the moon is at the horizon. If we see the "horizon moon" as more distant, we will perceive it as larger (Kaufman & Rock, 1989; Rock & Kaufman, 1962). This explanation is appealing, because it relates the moon illusion to our misperceptions of distance, a factor already accepted as important for many other illusions.

Many psychologists are not satisfied with this explanation, however, mostly because they are not convinced that the horizon moon looks farther away than the overhead moon. If we ask people which looks farther away, many say they are not sure. If we prevail upon them to answer, most say the horizon moon looks *closer*, in contradic-



FIGURE 4.64 Ordinarily, the moon looks much larger at the horizon than it does overhead. In photographs, this illusion disappears completely or almost completely, but the photographs do serve to demonstrate that the physical image of the moon is the same in both cases. The moon illusion requires a psychological explanation, not a physical one.

tion to the theory. Some psychologists reply that the situation is complicated: We unconsciously perceive the horizon as farther away; consequently, we perceive the horizon moon as very large; then, because of the perceived large size of the horizon moon, we secondarily and consciously perceive it as closer (as people report), although we continue to unconsciously perceive it as farther away (Rock & Kaufman, 1962).

One major message arises from work on optical illusions, and indeed from all the research on visual perception: What we perceive is not the same as what is “out there.” Our visual system does an amazing job of providing us with useful information about the world around us, but under unusual circumstances we can be very wrong about what we think we see.

THE MESSAGE Making Sense out of Sensory Information

You have probably heard the expression, “Seeing is believing.” The saying is true in many ways, including that what you believe influences what you see. Perception is not just a matter of adding up all the events striking the retina; we look for what we expect to see, we impose order on haphazard patterns, we see three dimensions in two-dimensional drawings, and we see optical illusions. Everything we have experienced in the past, everything that we know (or think that we know) influences what we perceive today.

S U M M A R Y

- ★ **Perception of minimal stimuli.** There is no sharp dividing line between sensory stimuli that can be perceived and sensory stimuli that cannot be perceived. (page 132)
- ★ **Signal detection.** To determine how accurately someone can detect a signal or how accurately a test diagnoses a condition, we need to consider not only the ratio of hits to misses when the stimulus is present, but also the ratio of false alarms to correct rejections when the stimulus is absent. (page 133)
- ★ **Subliminal perception.** Under some circumstances, a weak stimulus that we do not consciously identify can influence our behavior, at least weakly or briefly. However, the evidence does not support claims of powerful, irresistible effects. (page 134)
- ★ **Face recognition.** People are amazingly good at recognizing faces, partly by individual features such as eyes or nose, but also partly by the overall pattern, head size and shape, and the context. (page 135)
- ★ **Detection of simple visual features.** In the first stages of the process of perception, feature-detector cells identify lines, points, and simple movement. Feature detectors can-

not account for many of the active, interpretive aspects of perception, however. (page 137)

- ★ **Perception of organized wholes.** According to Gestalt psychologists, we perceive an organized whole by identifying similarities and continuous patterns across a large area of the visual field. (page 139)

- ★ **Visual constancies.** We ordinarily perceive the shape, size, and color of objects as constant, even though the pattern of light striking the retina varies from time to time. (page 143)

- ★ **Motion perception.** We perceive an object as moving if it moves relative to its background. We can generally distinguish between an object that is actually moving and a similar pattern of retinal stimulation that results from our own movement. (page 143)

- ★ **Depth perception.** To perceive depth, we use the retinal discrepancy between the views that our two eyes see. We also use other cues that are just as effective with one eye as with two. People need some experience with photos or drawings before they can use depth cues to interpret sizes and distances in the drawings. (page 144)

- ★ **Optical illusions.** Many, but not all, optical illusions result from interpreting a two-dimensional display as three-dimensional or from other faulty estimates of depth. (page 146)

- ★ **The size-distance relationship.** Our estimate of an object's size depends on our estimate of its distance from us. If we overestimate its distance, we will also overestimate its size. (page 147)

Suggestion for Further Reading

Rock, I. (1984). *Perception*. New York: Scientific American Books. Includes discussions of visual constancies, illusions, motion perception, and the relationship between perception and art.

Terms

- sensory threshold** the intensity at which a given individual can detect a sensory stimulus 50% of the time; a low threshold indicates the ability to detect faint stimuli (page 133)
- absolute threshold** the sensory threshold at a time of maximum dark adaptation (page 133)
- signal-detection theory** the study of people's tendencies to make hits, correct rejections, misses, and false alarms (page 133)
- subliminal perception** the ability of a stimulus to influence our behavior even when it is presented so faintly or briefly or along with such strong distractors that we do not perceive it consciously (page 134)
- feature detector** a neuron in the visual system of the brain that responds to the presence of a certain simple feature, such as a horizontal line (page 137)
- waterfall illusion** a phenomenon in which prolonged staring at a waterfall and then looking at nearby cliffs causes those cliffs to appear to flow upward (page 138)
- Gestalt psychology** an approach to psychology that seeks to explain how we perceive overall patterns (page 139)

figure and ground an object and its background (page 140)

reversible figure a stimulus that you can perceive in more than one way (page 140)

proximity in Gestalt psychology, the tendency to perceive objects that are close together as belonging to a group (page 140)

similarity in Gestalt psychology, the tendency to perceive objects that resemble each other as belonging to a group (page 141)

continuation in Gestalt psychology, the tendency to fill in the gaps in an interrupted line (page 141)

closure in Gestalt psychology, the tendency to imagine the rest of an incomplete, familiar figure (page 141)

good figure in Gestalt psychology, the tendency to perceive simple, symmetrical figures (page 141)

visual constancy the tendency to perceive objects as unchanging in shape, size, and color, despite variations in what actually reaches the retina (page 143)

induced movement a perception that an object is moving and the background is stationary when in fact the object is stationary and the background is moving (page 143)

stroboscopic movement an illusion of movement created by a rapid succession of stationary images (page 144)

phi effect the illusion of movement created when two or more stationary lights separated by a short distance flash on and off at regular intervals (page 144)

depth perception the perception of distance, which enables us to experience the world in three dimensions (page 144)

retinal disparity the difference in the apparent position of an object as seen by the left and right retinas (page 144)

convergence the degree to which the eyes turn in to focus on a close object (page 144)

binocular cues visual cues that depend on the action of both eyes (page 144)

monocular cues visual cues that are just as effective with one eye as with both (page 144)

motion parallax the apparently swift motion of objects close to a moving observer and the apparently slow motion of objects farther away (page 146)

optical illusion a misinterpretation of a visual stimulus as being larger or smaller, or straighter or more curved, than it really is (page 146)

visual capture effect the tendency to localize a sound as coming from a prominent visual feature (such as a loudspeaker or a ventriloquist's dummy) (page 149)

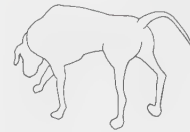
moon illusion the apparent difference between the size of the moon at the horizon and its size when viewed higher in the sky (page 150)

16. Object size is a cue that the elephant (usually larger than a deer) must be far away. Interposition is a cue in part a; the deer overlaps a hill that overlaps the hill with the elephant. Linear perspective is a cue in part b. (page 146)
17. Retinal disparity. (page 146)

Answers to Other Questions in the Text

C. a (7). b (1). c (5). d (9). e (4).

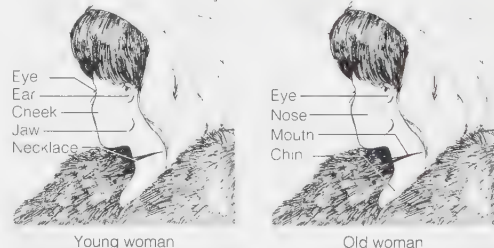
D.



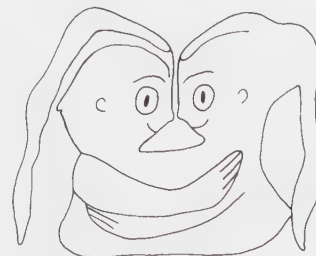
E.



F.



G.



- H. Line d is the same length as the one on the right. (Check it with a ruler.)

Answers to Concept Checks

14. We have been told the "hit" rate, but we cannot evaluate it unless we also know the "false alarm" rate. That is, how many people without any alcohol or drug problem have this same pattern of brain waves? If that percentage is large, the test is useless. The smaller that percentage is, the better. (page 134)
15. In Figure 4.41, continuation, closure, and perhaps good figure; in Figure 4.42, closure. (page 142)

Web Resources

Perception & Action Projects

george.arc.nasa.gov/PBA_Group/html/open.html

NASA has posted demonstrations of perceptual illusions and examples of behavioral adaptation by Malcolm M. Cohen.

Reverse Speech Home Page

www.reversespeech.com/

You've heard that particular recordings contain hidden messages that can be heard only if the material is played backwards.

One person had the time and patience to research such claims. You will need the RealAudio plug-in, available for free at www.realaudio.com/products/player/index.html. Can you provide a better explanation for the reversed messages?

Altered States



5

★ MODULE 5.1

Sleep and Dreams

Our Circadian Rhythms

Morning People and Evening People / Shifting Sleep Schedules / Brain Mechanisms of Circadian Rhythms

Why We Sleep

The Repair and Restoration Theory of Why We Sleep / The Evolutionary or Energy-Conservation Theory

Stages of Sleep

Sleep Cycles During the Night / Sleep Stages and Dreaming / The Functions of REM Sleep

The Content of Our Dreams

Abnormalities of Sleep

Insomnia / DISORDERS RELATED TO INSOMNIA: WHAT'S THE EVIDENCE? / Sleep Apnea / Sleep Talking, Sleepwalking, Nightmares, and Night Terrors / Leg Movements While Trying to Sleep / Hypersomnia / If You Have Trouble Sleeping . . .

★ MODULE 5.2

Hypnosis

Ways of Inducing Hypnosis

The Uses and Limitations of Hypnosis

What Hypnosis Can Do / What Hypnosis Cannot Do / THE LIMITS OF HYPNOSIS: WHAT'S THE EVIDENCE?

Distortions of Perception Under Hypnosis

Is Hypnosis an Altered State of Consciousness?

How Well Can an Unhypnotized Person Pretend to Be Hypnotized? / Faked Hypnosis by a Criminal Defendant

Meditation: In Some Ways like Hypnosis

★ MODULE 5.3

Drugs and Their Effects

A Survey of Abused Drugs and Their Effects

Alcohol / Tranquilizers / Opiates / Marijuana / Stimulant Drugs / Hallucinogens

Chronic Drug Effects

Drug Withdrawal / Drug Tolerance

During surgery, anesthetized patients are silent, move very little, and show no sign of pain. And if you were to ask them afterward whether they remember anything that happened or anything they heard during the surgery, they would insist that they do not. Nevertheless, if you were to provide a list of words and require them to guess which ones were read aloud during the operation, most of their guesses would be correct—if you ask them within 36 hours (Merikle, 1996).

So, were they conscious during the operation, even though they gave every indication of being unconscious? Or did they remember (weakly) certain events that occurred when they were unconscious? Evidently, the distinction between consciousness and unconsciousness is not a yes-or-no issue: There are various degrees of consciousness.

Consciousness is a fascinating topic, but it is also one of the most difficult phenomena to investigate (Cohen & Schooler, 1997). Does consciousness depend on certain brain areas or processes more than others? Is it really unitary, or do many brain areas each have their own “pieces” of consciousness? Which nonhuman animals (if any) are conscious? What about preverbal children? newborns? fetuses in the middle of pregnancy? brain-damaged people? computers?

Some psychologists regard these questions as answerable; others do not. The main problem is that consciousness, being an internal experience, can be observed by oneself but not by others. The best we can do scientifically is to examine procedures that seem to alter consciousness, such as sleep and dreams, hypnosis, and drugs.



From infancy to adulthood, you repeatedly add and replace material, as did this woman. Only a minuscule fraction of the molecules in your body today are the same as the molecules that were present in infancy. What makes you the same person from one year to the next is your continuous stream of conscious experience.

Sleep and Dreams

Why do we sleep?

What accounts for the content of our dreams?

Ground squirrels hibernate during the winter, a time when they would have trouble finding food. The females awaken in spring as soon as food becomes available. The males also need to awaken in time to eat, but they have another concern as well: The females are ready to mate as soon as they come out of their winter burrows, and each female mates only once a year. Any male who awakens later than the females pays for his extra rest by missing his only mating opportunity for the *entire year*. So males don't take any chances; they all awaken from hibernation a full week before the females do. And then they sit around waiting . . . with no females, nothing to eat, and little to do except to fight with one another (French, 1988).

The point is that animals have evolved internal timing mechanisms to prepare their behavior for predictable needs. Male ground squirrels have a mechanism that awakens them from hibernation while the air is still cold and well before food is available. They awaken not in response to their current situation but in preparation for what will happen a few days later. Similarly, most migrating birds start south in the fall well before their northern homes become inhospitable.

Humans also have built-in timing mechanisms. We do not have an annual mechanism for migration or hibernation, but we do have mechanisms to prepare us for activity during the day and for sleep during the night.

Our Circadian Rhythms

Humans and other highly visual animals are active during the day and inactive at night (Figure 5.1). Rats, mice, and other less visual animals are active at night and inactive during the day. Each species generates a **circadian rhythm**, a *rhythm of activity and inactivity lasting about one day*. (The term *circadian* comes from the Latin roots

circa and *dies*, meaning “about a day.”) The rising and setting of the sun provide cues to reset our rhythm each day; in an environment with no cues for time, most people will generate a waking-sleeping rhythm lasting a little longer than 24 hours (Moore-Ede, Czeisler, & Richardson, 1983).

One of the earliest demonstrations of humans' circadian rhythms was a study of two people who spent a few weeks in a remote part of Mammoth Caves in Kentucky (Kleitman, 1963). For 24 hours per day, the temperature was a constant 12° Celsius, and the relative humidity was a steady 100%. They saw no light except the light from lamps that they controlled themselves, and they heard no noises except their own. Any decisions about when to sleep and when to awaken had to come from within, not from the environment. Yet, they went to sleep and awoke at about the same time every day.

Sleepiness and alertness depend strongly on one's position within the circadian rhythm. If you have ever gone all night without sleep—as most college students do at one time or another—you probably grew very sleepy by about 4:00 or 5:00 A.M. But if you were still awake at 7:00 or 8:00 A.M., you began feeling less sleepy, not more so. You became more alert because of your circadian rhythm, even though your sleep deprivation had continued.

In one study, volunteers went without sleep for three nights; an experimenter periodically took their temperature and measured their performance on logical reasoning tasks. Both temperature and logical reasoning declined during the first night and then increased the next morning. During the

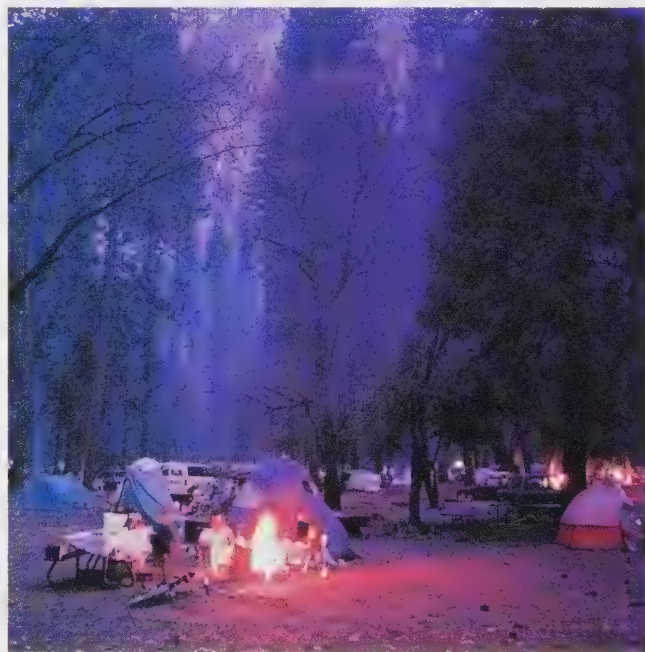


FIGURE 5.1 The rising and setting of the sun does not directly produce our daily rhythm of wakefulness and sleepiness, but it synchronizes that rhythm. We adjust our internally generated cycles so that we feel alert during the day and sleepy at night.

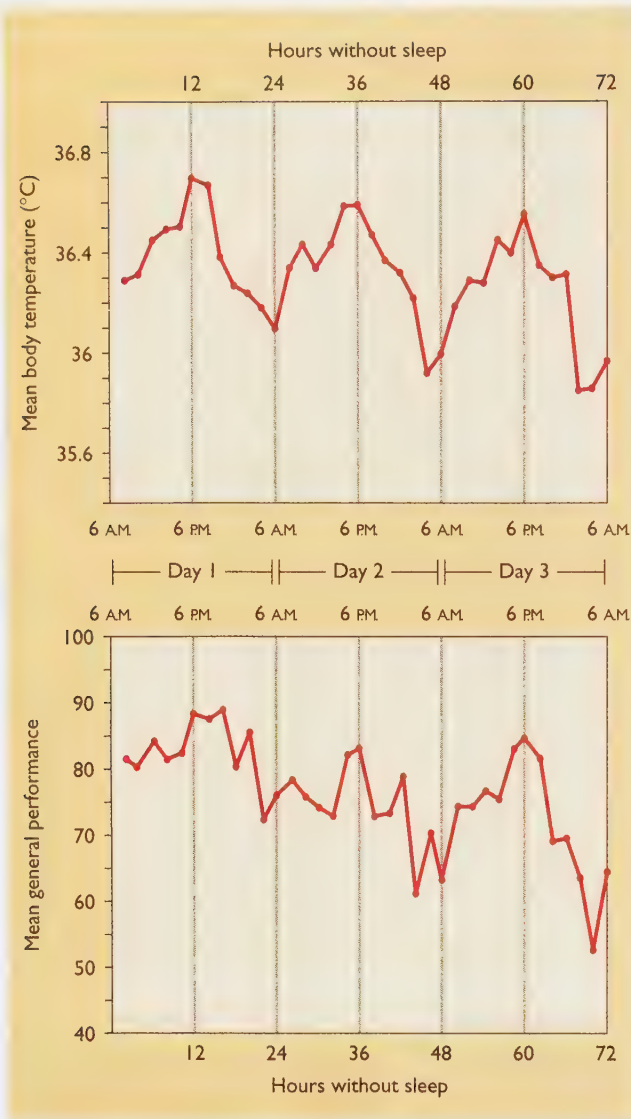


FIGURE 5.2 Cumulative effects of 3 nights without sleep: Both body temperature and logical reasoning decrease each night and increase the next morning. They also deteriorate from one day to the next. (From Babkoff, Caspy, Mikulincer, & Sing, 1991.)

second and third nights, temperature and reasoning decreased more than they had the first night, but they again improved the following morning (Figure 5.2). Thus, sleep deprivation produces a pattern of progressive deterioration that is superimposed on the normal circadian cycle of rising and falling body temperature and alertness (Babkoff, Caspy, Mikulincer, & Sing, 1991). In short, sleepiness apparently depends partly on how long one has gone without sleep and partly on the time of day (that is, where one is within the circadian rhythm).

Morning People and Evening People

Our circadian rhythms do not affect us all in exactly the same way. “Morning people” awaken early and full of energy,

doing their best work before noon; “evening people” take longer to become active in the morning and feel best and do their best work in the late afternoon or evening. Most people can diagnose themselves correctly as either morning or evening types. One study found that self-described “morning people” reach their peak body temperature as well as their intellectual and physical peak in the mornings. People who call themselves “evening people” are quite literally slow to warm up in the mornings; their body temperatures are low in the mornings and rise gradually during the day. They also reach their intellectual and physical peak between 5:00 and 10:00 P.M. (Horne, Brass, & Pettitt, 1980).

The morning person/evening person distinction is particularly relevant for research on aging. Most young people are either neutral (about equally alert at all times) or “evening people,” whereas nearly all people over age 65 are “morning people.” For many years, researchers reported substantial declines in memory with older people. But who did most of this research? Graduate students. How old are graduate students? Most are in their 20s. What time do graduate students choose to conduct their research? Late afternoon or evening, a fine time for 25-year-olds but a miserable time for 65-year-olds. Researchers in one study compared the memories of young adults (18–22 years old) and older adults (66–78 years old) at different times of day. Early in the morning, the older adults did just as well as the younger ones. However, as the day went on, the younger adults either remained steady or improved; the older adults deteriorated steadily (May, Hasher, & Stoltzfus, 1993). Figure 5.3 shows the results of this study.

SOMETHING TO THINK ABOUT

Are most college classes offered in the early morning or late afternoon? Is that because *you* want to take them at that time, or because your aging professors want to schedule them then? ★

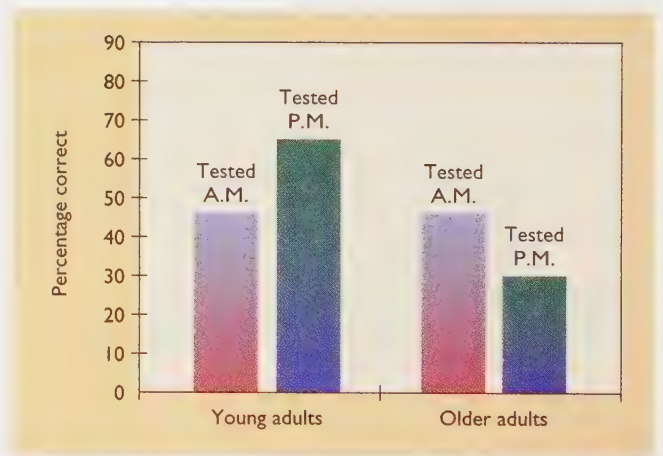


FIGURE 5.3 If tested early in the morning, older people perform as well as younger people on memory tasks. As the day progresses, young people improve and older people deteriorate.

Shifting Sleep Schedules

In ordinary life, the light of early morning serves to reset the body's clock each day to prevent it from gradually slipping out of synchrony with the outside world. If you travel across time zones, the light in your new location will eventually reset your clock to the new time, but until it does so, your internal rhythms will be out of phase with the light and dark cycles of your new environment. For example, if you travel from California to France, it will be 7:00 A.M. (time to get up) when your body says it is 10:00 P.M. (not quite time to go to bed). You will experience **jet lag**, a *period of discomfort and inefficiency, because your internal clock is out of phase with your new surroundings*. You will gradually adjust to the new schedule, although some people adjust much faster than others.

Most people find it easier to adjust when flying west, where they go to bed later, than flying east, where they go to bed earlier (Désir et al., 1981). Thus, east-coast people adjust to west-coast time more easily than west-coast people adjust to east-coast time (Figure 5.4). International pilots, who must constantly adjust to one schedule and then another, develop a variety of techniques to combat fatigue—such as napping, conserving energy, and drinking coffee (Petrie & Dawson, 1997).

It is possible to have an experience akin to jet lag without leaving town. Suppose you stay up late on Friday night and wake up late on Saturday morning. (Many readers will not find this difficult to imagine.) Then, you stay up late again on Saturday and wake up late on Sunday. By Monday morning, when it is time to awaken for school or work, your circadian rhythm has readjusted, so even though the clock on your table says 7:00 A.M., your internal clock thinks you are somewhere far to the west where the time is only 5:00 A.M.

Companies that want to keep their factories going non-stop run three work shifts, such as midnight–8:00 A.M., 8:00 A.M.–4:00 P.M., and 4:00 P.M.–midnight. Because few

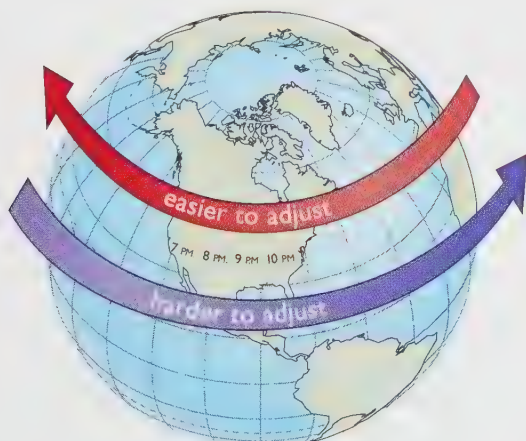


FIGURE 5.4 People traveling east suffer more serious jet lag than people traveling west.

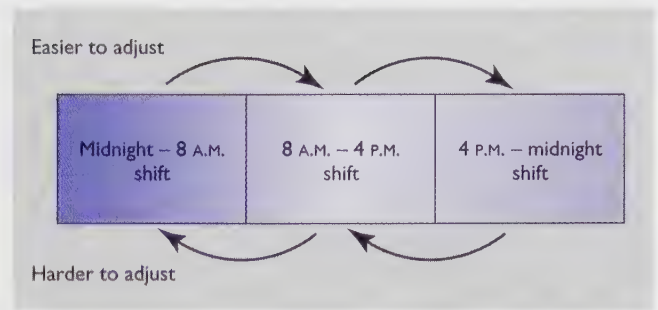


FIGURE 5.5 The graveyard shift is aptly named—serious industrial accidents usually occur at night, when workers are least alert. Night-shift jobs providing emergency services are essential. But few people want to work permanently at night, so workers rotate among three shifts. As in jet lag, the direction of change is critical. Moving forward—clockwise—is easier than going backward.

people want to work regularly on the “graveyard shift” (midnight–8:00 A.M.), many companies rotate their workers among the three shifts. Suppose that you worked the 8:00 A.M.–4:00 P.M. shift today but tomorrow you are scheduled to work the midnight–8:00 A.M. shift. You can’t get to sleep at 4:00 in the afternoon, so you show up for work without having slept. You are likely to make many mistakes during the night, and by the time you are ready to drive home the next morning, your coordination and judgment will be as badly impaired as if you were legally drunk (Dawson & Reid, 1997). Even people who work the night shift month after month may continue feeling groggy on the job and sleeping fitfully during the day.

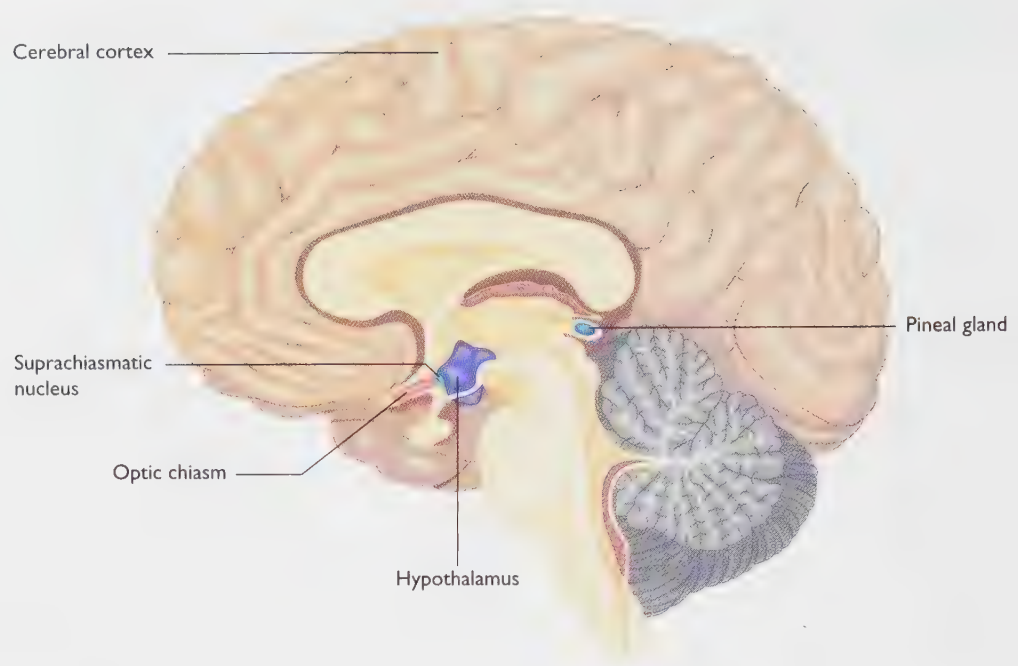
There are two ways for employers to ease the burden on their workers: First, when they transfer workers from one shift to another, they should transfer workers to a *later* shift, not an earlier shift (Czeisler, Moore-Ede, & Coleman, 1982) (Figure 5.5). That is, someone working the 8:00 A.M.–4:00 P.M. shift should switch to the 4:00 P.M.–midnight shift (equivalent to traveling west), instead of the midnight–8:00 A.M. shift (equivalent to traveling east).

Second, employers can help workers adjust to the night shift by providing bright lights to mimic sunlight. In one study, young men exposed to very bright lights at night adjusted well to working at night and sleeping during the day. Within six days, their circadian rhythms had shifted to the new schedule. Another group of men who worked on the same schedule but under dimmer lights showed no indications of altering their circadian rhythms (Czeisler et al., 1990).

CONCEPT CHECK

1. Suppose you are the president of Consolidated Generic Products in the United States. You are negotiating a difficult business deal with someone from

FIGURE 5.6 The suprachiasmatic nucleus, a small area at the base of the brain, produces the circadian rhythm. Ordinarily, information about light intensity reaches this area via the optic nerves from the eyes; such information resets the circadian rhythm but is not necessary for its generation. Cells in the suprachiasmatic nucleus can generate a 24- to 25-hour rhythm of activity on their own, even if they are separated from the rest of the brain.



the opposite side of the world. Should you prefer a meeting place in Europe or on an island in the Pacific Ocean? (Check your answer on page 172.)

SOMETHING TO THINK ABOUT

Imagine someone who cannot get to sleep until 3:00 A.M.; for months she has been trying to get to bed earlier, but failing. What advice would you offer? (Hint: Remember that our internal clocks can shift more easily to a later time than to an earlier time.) ★

Brain Mechanisms of Circadian Rhythms

The circadian rhythm of sleep and wakefulness is generated by a tiny structure at the base of the brain, known as the *suprachiasmatic nucleus*. If that area of the brain is damaged, the body's activity cycles become erratic (Rusak, 1977); if cells from that area are kept alive outside the body, they generate a circadian rhythm on their own (Green & Gillette, 1982; Inouye & Kawamura, 1979). In short, this area of the brain serves as the body's built-in clock. (See Figure 5.6.)

The suprachiasmatic nucleus exerts its control partly by regulating secretions of the hormone *melatonin* by the pineal gland. Ordinarily, the human pineal gland starts releasing increased amounts of melatonin at about 8:00 to 10:00 P.M., so people find it easy to fall asleep about two or three hours later. If you take a melatonin pill in the evening, you will notice little if any effect, because you were already producing increased melatonin at that time. However, if you have just flown a few time zones east and want to get to

bed two or three hours before your biological clock would ordinarily allow, then a melatonin pill can be very helpful indeed (Deacon & Arendt, 1996).

CONCEPT CHECK

- Suppose you are required to work the midnight–8:00 A.M. shift, and you would like to go to sleep at 4:00 P.M. in order to be well-rested before starting work. Would a melatonin pill help? If so, when should you take it? (Check your answer on page 172.)

Why We Sleep

We would not have been born with a mechanism that forces us to sleep for 8 hours or so out of every 24 unless sleep did us some good. But what good does sleep do? Scientists have proposed two theories.

The Repair and Restoration Theory of Why We Sleep

According to the **repair and restoration theory**, the *purpose of sleep is to enable the body to recover from the exertions of the day*. We undeniably need sleep, and we suffer if we do not get it. Sleep deprivation leads to impaired concentration and irritability; prolonged deprivation can impair the function of the immune system and thus increase the risk of becoming ill (Dement, 1972; Everson, 1995; Rechtschaffen & Bergmann, 1995). However, most of the restorative functions that occur during sleep—such as digestion